

# Rail Track Temperature Monitoring System

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**A**ll Indian Railway Permanent Way offices monitor rail tracks to record their maximum and minimum temperature of the day at hourly intervals. It is done to check the metal shrinkage during winters and expansion during hot weather. This is required for track maintenance activities such as hot or cold weather patrolling, packing of gaps found in the tracks, checking long welded rail (LWR) tracks, and other issues that could cause derailment and mishaps.

This GSM based device will keep the concerned officer informed about the track temperature on his mobile phone with a text message from a designated SIM card in the circuit on hourly basis. The author's prototype is shown in Fig. 1.

## Circuit and working

Circuit diagram of the rail track temperature monitoring system with hourly updates is shown in Fig. 2. It comprises an Arduino Nano board, an A6 GSM module, and a DS18B20 temperature sensor.

The circuit can be powered by any 5-volt, 2-amp regulated DC power supply connected across CON1. The 5-volt supply thus goes to Vin pin of Arduino Nano and Vcc pins of GSM module and DS18B20 sensor.

Pins uTX and uRX of the GSM module are connected to pins D3 and D2 of Arduino Nano, respectively. Both of its ground pins are connected to GND pin of the power supply.

The temperature sensor (DS18B20) has three pins, one of which is connected to 5-volt supply, another to GND pin, and the third to pin D4 of Arduino. The 4.7-kilo-ohm resistor R1 works as pull-up resistor.

The GSM module needs an antenna to get proper signal strength to

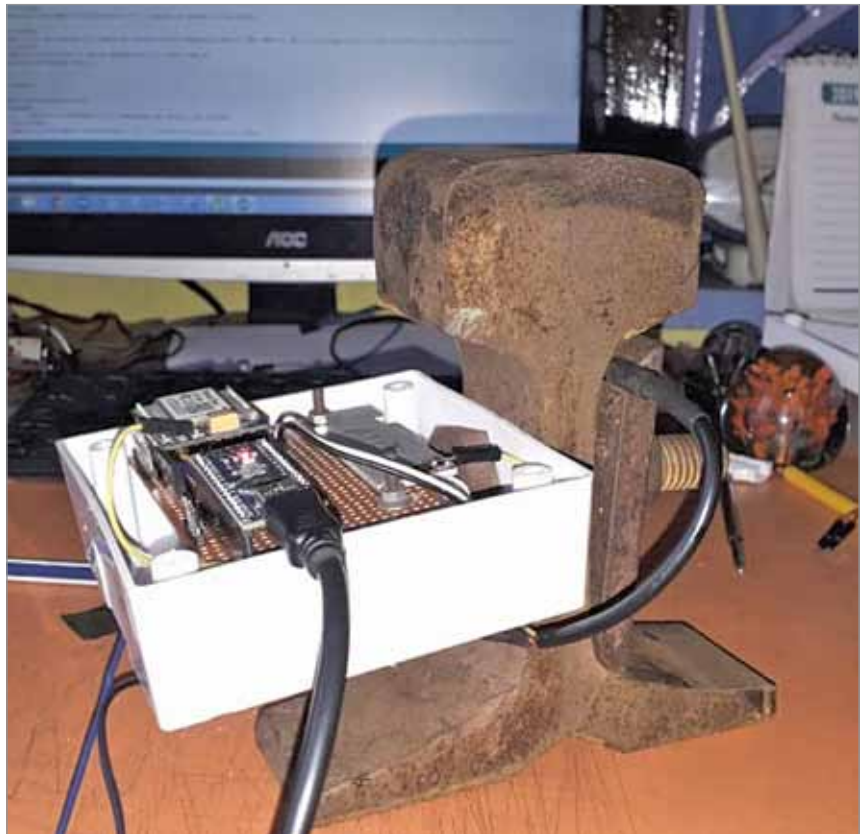


Fig. 1: Author's prototype

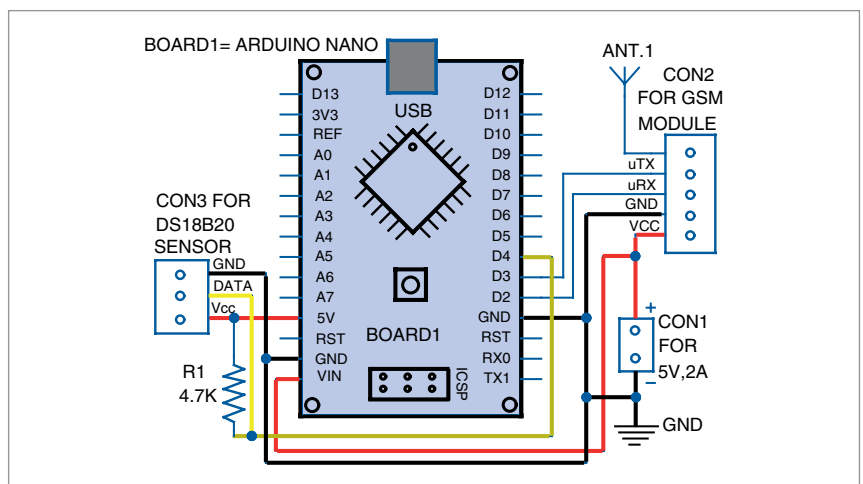


Fig. 2: Circuit diagram of the monitoring system